

Calligraphy Experience System That Conveys the Relationship Between Kanji and Nature

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ABSTRACT

To date, systems dealing with Kanji characters have been produced mainly for the purpose of learning, but it is important not only to learn the meaning and writing order of Kanji characters, but also to understand the cultural aspects of them. In regions where Kanji characters are used, various cultures, including Kanji characters, are influenced by nature, so in this study, we propose a calligraphy experience system that conveys the relationship between nature and Kanji characters. When a user looks at the writing order projected on the tabletop and writes Kanji characters created from natural shapes with a brush, the written characters change into the original natural shapes. The user's handwriting could be transformed into a natural form by using projected images.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Mixed/augmented reality**;

KEYWORDS

Culture, Calligraphy, Interaction, XR

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1 INTRODUCTION

In our research to date, we have taken traditional cultures such as shrines [Kobayashi and Hoshino, 2020], furoshiki (wrapping cloth) [Kobayashi et al., 2019], and Japanese food [Kobayashi et al., 2021] as subjects, and created a system that recognizes the user's behavior through sensors and other devices during actual cultural experiences, and projects appropriate information on the tabletop based on the user's behavior. For this reason, we looked at writing from a cultural perspective and focused on hieroglyphics among Kanji characters, which were created based on natural shapes. This is because shrines in Japan, one of the countries where Kanji characters are used, enshrine natural objects such as mountains, rocks, trees,

and waterfalls as objects where gods reside. In addition, Japanese food is made with the original taste of seasonal ingredients. In this study, we propose a calligraphy experience system in which a user writes Kanji characters created from natural shapes with a brush while looking at the writing order projected on a tabletop, and the written characters change into the original natural shapes.

2 RELATED STUDIES

For games aimed at memorizing the order of Kanji characters, music is incorporated to enhance the immersive learning experience [Keehl and Melcer, 2019]. Adventure-style serious games also incorporated turn-based combat to learn basic Kanji characters [Plecher et al., 2018]. Previous studies have been for learning purposes and have not utilized cultural experiences. In the calligraphy experience, it is easy to connect the experience with information and effectively convey the relationship between Kanji characters and nature.

3 SYSTEM CONFIGURATION

The system consists of a PC, a camera, a projector, two VIVE base stations, a Tundra tracker, and a brush. The software was created using Unity. The camera and base stations were set up with light stands, and the short focus projector was fixed with fittings for tabletop projection.

Since the position of the brush needs to be measured in order to express the user's handwriting on the projected image of the projector, a tracker was attached to the brush. From the position and rotation of the tracker and the length of the brush, contact with the paper can be recognized at any brush angle by estimating the contact of the brush tip. When the tip of the brush is slightly touching the paper, the line is made thin, and when the brush is firmly touching the paper, the line is made thick.

Considering that the type and number of Kanji characters to be included in the system are easy enough for children and those who do not know Kanji to easily experience them, and that the experience time per person is not too long, eight characters were selected: rice field (田), mountain (山), river (川), tree (木), fire (火), sun (日), rain (雨), and moon (月) (Figure 1). Each of the eight papers for each Kanji character is accompanied by a marker and a model of the character to be written, and when it is placed on the table, a camera recognizes the type of Kanji character from the marker and the writing order is projected to the center of the tabletop. When the user finishes writing while looking at the writing order projected on the tabletop, an animation is projected in which the writing changes to its original natural form. If you write "tree (木)," for example, your handwriting will gradually change into the actual shape of a tree (Figure 2). The animation was created in handwriting that had an affinity with writing with a brush.

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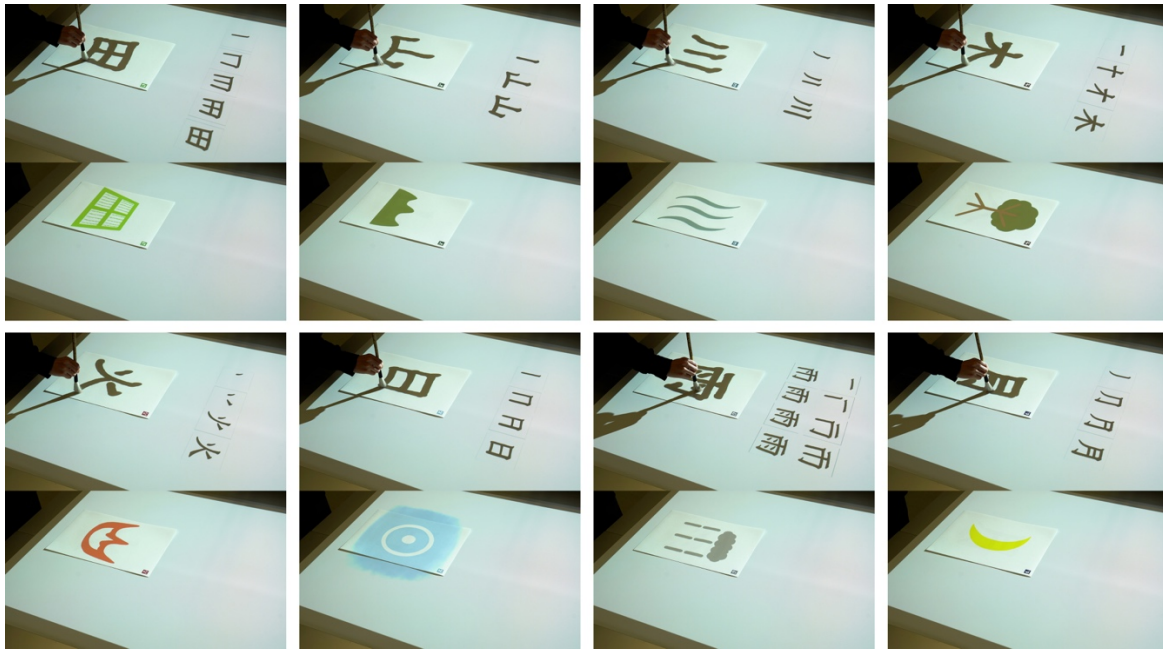


Figure 1: Eight Kanji Characters (From top left: rice field (田), mountain (山), river (川), tree (木), fire (火), sun (日), rain (雨), and moon (月))

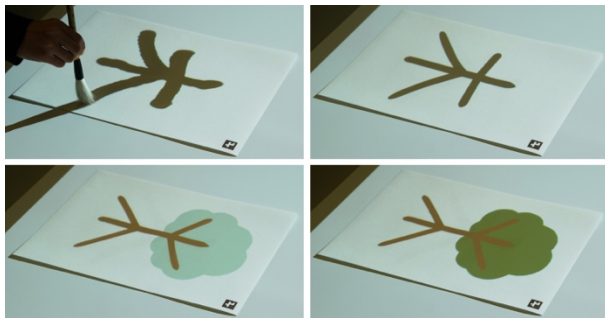


Figure 2: Kanji Changes to its Original Natural Shape

Character recognition is performed by judging whether the model area displayed on the paper is filled in. Since there is a wide range in the thickness of the characters written by the user, the system recognizes the central part of the handwritten character that the user's writing line must pass through.

As soon as the character change is finished, the changed character appears on the other side of the user as part of the landscape of the ink painting, and the landscape painting is completed by writing all the Kanji characters (Figure 3). The feature was intended to emphasize and convey that the formation of characters is influenced by nature and to promote the cultural experience of the users.

4 CONCLUSION

In this study, in order to convey the relationship between Kanji characters and nature, we proposed a calligraphy experience system

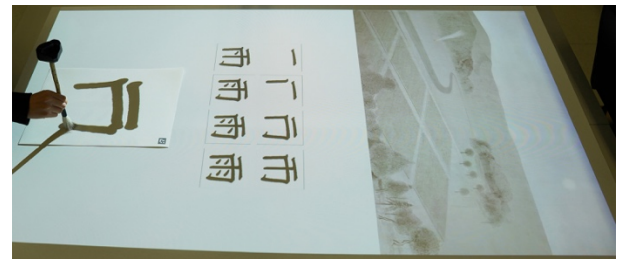


Figure 3: Landscape of the Ink Painting

in which a user writes Kanji characters made from natural shapes with a brush, and the written characters change into the original natural shapes. Future prospects include improving the system by allowing multiple people to experience it.

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